

Total marks (84)

Attempt questions 1 – 7

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 marks)

Marks

- (a) Find: $\frac{d}{dx} \tan^{-1}(2x)$ 1
- (b) A and B are the points $(-5, 12)$ and $(4, 9)$ respectively. P is the point which divides AB internally in the ratio $3 : 2$. Find the coordinates of P . 2
- (c) Evaluate: $\lim_{x \rightarrow 0} \left(\frac{\tan 3x}{2x} \right)$ 1
- (d) Find the acute angle, in degrees correct to one decimal place, between the two curves $y = x^2$ and $y = x$ at the point of intersection $(1, 1)$. 2
- (e) Find: $\int \sin^2 3x \, dx$ 2
- (f) Using the substitution $u = x - 1$, evaluate $\int_2^5 \frac{x}{\sqrt{x-1}} \, dx$. 4

- (a) Let α, β and γ be the roots of the equation $x^3 - 5x^2 - 2x - 8 = 0$.

Without finding the actual roots, evaluate:

(i) $\alpha + \beta + \gamma$ 1

(ii) $\alpha\beta + \alpha\gamma + \beta\gamma$ 1

✓ (iii) $\alpha^2 + \beta^2 + \gamma^2$ 2

- (b) Let $f(x) = \ln x - \sin x$. It is known that the real root of $f(x) = 0$ lies between $x = 2$ and $x = 2.5$. 3

Using one application of the *'halving the interval'* method, determine whether the root of $f(x) = 0$ is closer to $x = 2$ or $x = 2.5$.

- ✓ (c) Find the volume of the solid generated when the area under the curve $y = \frac{1}{(1-9x^2)^{\frac{1}{4}}}$, above the x -axis and between $x = 0$ and $x = \frac{1}{3\sqrt{2}}$, is rotated about the x -axis. 3

- (d) ✓ (i) Write down the value of the constant k in the equation $5^x = e^{kx}$, $x \neq 0$. 1

(ii) Hence or otherwise, find $\frac{d}{dx}(5^x)$. 1

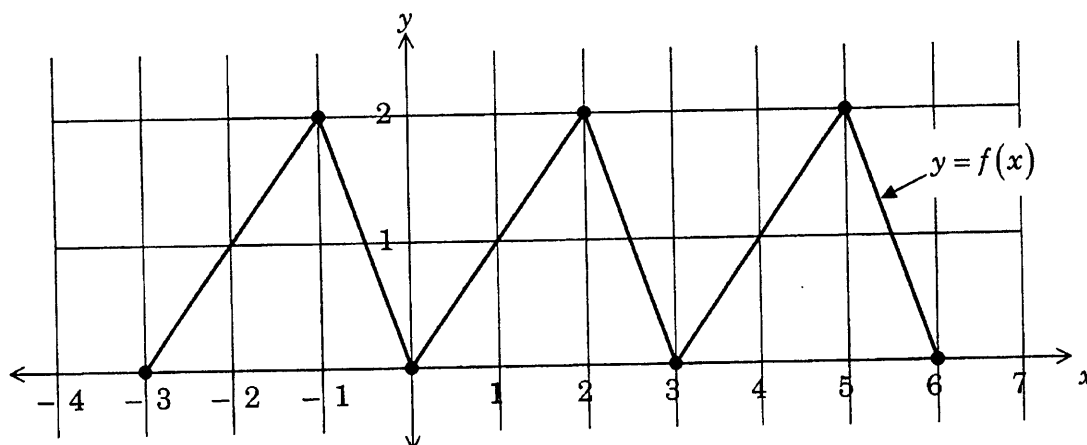
- ✓ (a) Solve $2\sin x + \cos x = -1$, for $0 \leq x \leq 2\pi$, by first using the substitution, $t = \tan \frac{x}{2}$. 3

- (b) Prove by mathematical induction that if n is a positive integer, then: 3

$$\frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \frac{1}{9 \times 13} + \dots + \frac{1}{(4n-3)(4n+1)} = \frac{n}{4n+1}$$

- ✓ (c) Without using a calculator, find the exact value of $\sin\left(\cos^{-1}\frac{2}{3} + \sin^{-1}\frac{1}{4}\right)$. 3

(d)



The diagram above shows the graph of a periodic function $y = f(x)$ over the interval $-3 \leq x \leq 6$.

- (i) State the period of $y = f(x)$. 1
- (ii) Assuming that the period of the function $y = f(x)$ continues to have the same form over the interval $-30 \leq x \leq 60$, calculate $f(52)$. 1
- (iii) Find $f'(x)$, when $x = 26\frac{1}{2}$. 1

- (a) Consider the function defined by $f(x) = x\left(\sqrt[3]{x^2 - 4}\right)$, where x is any real number, $f'(x) = \frac{5x^2 - 12}{3(x^2 - 4)^{\frac{2}{3}}}$ and $\left(2\sqrt{\frac{3}{5}}, -\frac{4\sqrt{3}}{5^{\frac{5}{6}}}\right)$ is one of the two stationary points on $y = f(x)$. *You do not need to verify these facts.*

- (i) Show that $f(x)$ is an odd function. 1
- (ii) Write down the coordinates of the second stationary point. 1
- (iii) Explain why there is a vertical tangent at $x = 2$. 1
- (iv) Sketch the graph of $y = f(x)$ and label the axes appropriately. 2

- (b) The function f is given by $f(x) = \cos^{-1}\left(\frac{x}{3}\right)$.

- (i) Find $f^{-1}(x)$. 1
- (ii) Write down the domain and range of $f^{-1}(x)$. 2
- (iii) Sketch the graph of $y = f^{-1}(x)$ and label the axes appropriately. 1



A particle is moving in simple harmonic motion about the point O . The point A , as shown in the diagram, is 8 metres from O . When the particle passes through the point A its speed is 3 ms^{-1} . The amplitude of the motion is 10 m.

- (i) Calculate the period of the motion. 2
- (ii) If x is the displacement of the particle from O , find the values of x for which the speed is zero. 1

Question 5 (12 marks)

Use a SEPARATE writing booklet

Marks

- (a) The acceleration of a particle is given by $\ddot{x} = 4(1+x)$, where x is the particle's displacement from the origin. The particle is initially at the origin with a velocity of 2 m/s. Let $v = \frac{dx}{dt}$.

(i) Prove that $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{d^2 x}{dt^2}$. 2

(ii) Find an expression for v in terms of x . 2

(i) Show that $x = e^{2t} - 1$. Note that when $t \geq 0$, $v > 0$. 2

- (b) One hundred grams of cane sugar in water are being converted into dextrose at a rate which is proportional to the amount unconverted at any time t , that is, if M grams are converted in t minutes, then,

$$\frac{dM}{dt} = k(100 - M), \text{ where } k \text{ is a constant}$$

(i) Verify that $M = 100 + Ae^{-kt}$, where A is a constant, satisfies the given differential equation. 2

(ii) If 40 grams are converted in the first 10 minutes, find A and k . 2

(iii) How many grams are converted in the first 45 minutes, correct to the nearest whole gram? 2

Question 6 commences on the next page

(a) Solve: $\frac{x}{x-1} \geq 5$

3

(b)

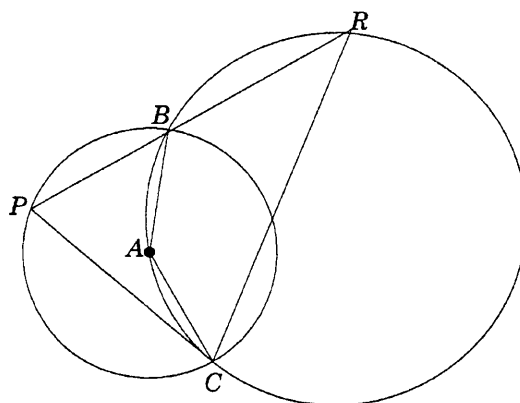


Diagram is not
to scale

A is the centre of the circle BCP. The point A lies on another circle BAC. The two circles intersect in B and C as shown in the diagram. PBR is a straight line.

Copy or trace this diagram into your writing booklet.

Prove, with reasons, that $RP = RC$.

3

(c) Two tangents from the external point $T(x_0, y_0)$ touch the parabola $x^2 = 4ay$ at $P(x_1, y_1)$ and $Q(x_2, y_2)$ respectively.

(i) Write down the Cartesian equation of the chord of contact in terms of x_0 and y_0 .

1

(ii) Show that the x values of the coordinates of P and Q are given by the roots of the equation $x^2 - 2x_0x + 4ay_0 = 0$.

2

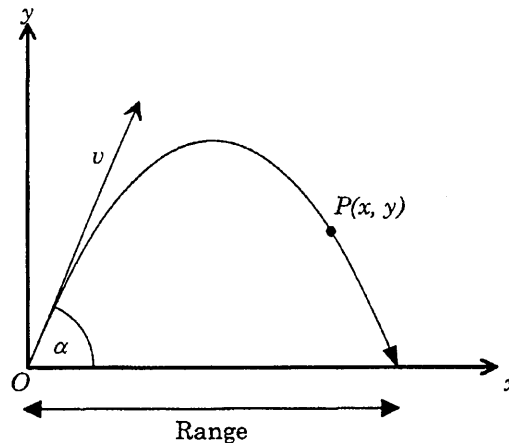
(iii) Show that the midpoint M of QP is given by $\left(x_0, \frac{x_0^2}{2a} - y_0\right)$.

2

(iv) Find the Cartesian equation of the locus of M .

1

(a)



A projectile is fired from level ground with an initial velocity, v metres per second, at an angle α to the horizontal. The origin, O , is taken to be at the point of projection on level ground.

- (i) Starting with $\ddot{x} = 0$, $\ddot{y} = -g$ and integrating, derive the parametric equations for the position of the projectile $P(x, y)$, after t seconds. 3
Ignore air resistance and assume the acceleration due to gravity is $g \text{ m/s}^2$.
- (ii) Prove that the horizontal range of the projectile from the point of projection, in metres, is given by $x = \frac{v^2 \sin 2\alpha}{g}$. 2
- (iii) A golf ball is driven with a velocity of 50 m/s at an angle α to the horizontal towards the hole on the green 250 metres away on the same horizontal plane as the point of projection. 2

At what angle should the golf ball be projected in order to achieve a 'hole-in-one', that is without bouncing or rolling first? Take $g = 9.8 \text{ m/s}^2$ and ignore air resistance.

Question 7 continues on the next page

Question 7 continued:

(b)

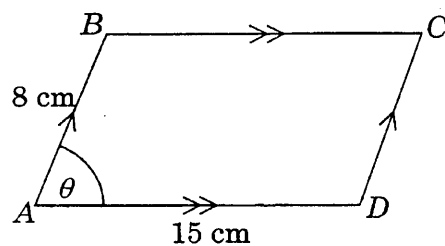


Diagram is not
to scale

A parallelogram $ABCD$ has initially sides of length 8 cm and 15 cm. 3

The angle θ at one of the vertices is decreasing at the rate of $\frac{\pi}{60}$ radians per minute.

Calculate the rate at which the area of the parallelogram is changing when $\theta = \frac{\pi}{6}$. Assume that as θ decreases, $ABCD$ remains a parallelogram.

- (c) Gambler buys three tickets in a lottery for which sixty tickets are sold in all. There will be five prizes awarded. Tickets drawn will not be replaced. 2

Find the probability that Gambler wins at least one prize.

End of Paper

QUESTION 6: (12 MARKS)

(a)

$$\frac{x}{x-1} \approx 7,5$$

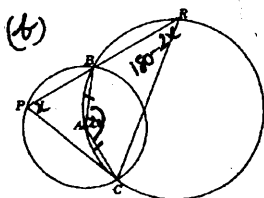
$$\left(\frac{x}{x-1}\right)(x-1)^2 \geq 5(x-1)^2$$

$$x(x-1) - 5(x-1)^2 \geq 0$$

$$(x-1)[x-5(x-1)] > 0$$

$$(x-1)(5-4x) \geq 0$$

$$\therefore \{x: 1 < x \leq \frac{5}{4}\}$$


$$AB = AC \text{ (equal radii)}$$

$$\angle BAC = 2x^\circ$$

$$\therefore \angle BPC = x^\circ$$

$\therefore \angle BPC \equiv 2$
(angle at the centre of
circle ~~is~~ is twice the
angle subt. on the same
arc).

Q6 ctd:

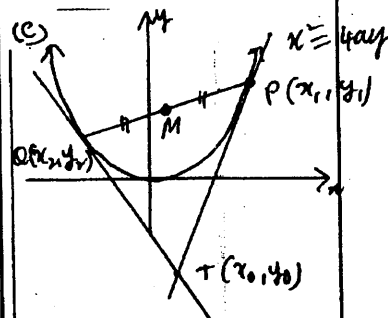
$\angle BRL = 180 - 2x$
opp. $\angle$ of a cyclic quad.
ABRL are supplementary.

$$\begin{aligned}\therefore \angle PCR &= 180 - (180 - 2x + x) \\ &= 180 - (180 - x) \\ &= x\end{aligned}$$

$\therefore \Delta PRC$ is isosceles

$$\therefore R_P = R_C$$

Consider opp equal Δ 's
in a trapezoid



(ii) Chord PQ:

$$\pi x_0 = 2a(y + y_0)$$

(ii) Solving $x \leq 4$ ay

$$\pi x_0 = 2a(y + y_0)$$

in ① $y = \frac{x^2}{4a}$

$$\therefore \text{in (2) } x x_0 = 2a \left(\frac{x^2}{4a} + y_0 \right)$$

$$\underline{16} \quad x x_0 = \frac{x^2}{2} + 2ay_0$$

$$e' 2x x_0 = x^2 + 4ay_0$$

$$x^2 - 2xx_0 + 4ay_0 = 0 \quad (3)$$

(iii) Solving (3)

$$x = \frac{2x_0 \pm \sqrt{(2x_0)^2 - 4(1)(4ay_0)}}{2}$$

$$= x_0 \pm \sqrt{x_0^2 - 4ay_0} \quad \checkmark$$

$$\begin{aligned} \text{let } x_1 &= x_0 + \sqrt{x_0^2 - 4ay_0} \\ x_2 &= x_0 - \sqrt{x_0^2 - 4ay_0} \end{aligned}$$

$$\therefore \frac{x_1 + x_2}{2} = \frac{2x_0}{2} = x_0 \quad (1)$$

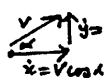
Subs ④ into $xx_0 = 2a(y+y_0)$

$$\therefore \frac{x_0^2 - 2ax_0}{2a} = y$$

$$\therefore y = \frac{x^2}{2a} - y_0$$

$$\therefore M = \left(x_0, \frac{x_0^2}{2a} - y_0 \right) \quad (5)$$

(iv) $x = x_0$ & $y = \frac{x_0^2}{2} - y_0$ from (5) ✓
 $\therefore y = \frac{x^2}{2a} - y_0 \Rightarrow 2a(y + y_0) = x^2$

Suggested Solution (a)	Comments	Suggested Solution (a)	Comments
<u>QUESTION 7: (12 MARKS)</u> (a) (i) $\ddot{x} = 0$  $\therefore \dot{x} = C_1$ when $t = 0$, $\dot{x} = V \cos \alpha = C_1$ $\therefore \dot{x} = V \cos \alpha$ $\therefore x = Vt \cos \alpha + C_2$ when $t = 0$, $x = 0$ $\therefore C_2 = 0$ $\therefore x = Vt \cos \alpha$ ① $\ddot{y} = -g$ $\therefore \dot{y} = -gt + C_3$ when $t = 0$, $\dot{y} = V \sin \alpha$ $\therefore \dot{y} = -gt + V \sin \alpha$ $y = -\frac{gt^2}{2} + Vt \sin \alpha + C_4$ when $t = 0$, $y = 0 \therefore C_4 = 0$ $\therefore y = -\frac{gt^2}{2} + Vt \sin \alpha$ ② $\therefore P(x, y)$ $= [Vt \cos \alpha, Vt \sin \alpha - \frac{gt^2}{2}]$ (ii), when $y = 0$, the particle has reached the ground or is on the ground to start with. Solving for $y = 0$ in ② $\frac{gt^2}{2} - Vt \sin \alpha = 0$	✓ ✓ ✓ ✓	$\therefore gt^2 - 2Vt \sin \alpha = 0$ $\therefore t[gt - 2V \sin \alpha] = 0$ $\therefore$ either $t = 0$ or $t = \frac{2V \sin \alpha}{g}$ — ③ Sub ③ into ① $x = V \left[\frac{2V \sin \alpha}{g} \right] (\cos \alpha)$ $= \frac{2V^2 \sin \alpha \cos \alpha}{g}$ $\therefore x = \frac{V^2 \sin 2\alpha}{g}$ as required. ④ (iii) From ④, $g = 9.8$, $x = 250$, $V = 50$. $250 = \frac{50^2}{9.8} \sin 2\alpha$ $\therefore \sin 2\alpha = 0.98$ $\therefore 2\alpha \doteq 78.52^\circ \dots$ $\alpha \doteq 39.26^\circ \dots$ $\therefore 2\alpha = \sin^{-1}(0.98)$ $\therefore \alpha = \frac{1}{2} \sin^{-1}(0.98)$ (in <u>exact</u> form)	✓ ✓ ✓ ✓ ✓ or equivalent

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
Q7 ctd.: (4), Area of a parallelogram $A = ab \sin \theta$ where a, b are the adjacent sides including θ. $\therefore A = 8 \times 15 \sin \theta$ $A = 120 \sin \theta$	✓	(c) $P(\text{win at least one prize})$ $= 1 - P(\text{win no prize})$ $= 1 - \left[\frac{57}{60} \times \frac{56}{59} \times \frac{55}{58} \times \frac{54}{57} \times \frac{53}{56} \right]$ $= \frac{1597}{6844} \quad (= 0.23)$	✓
Given $\frac{d\theta}{dt} = -\pi/60$ rad/min We want: $\frac{dA}{dt}$ $\frac{dA}{dt} = \frac{dA}{d\theta} \times \frac{d\theta}{dt}$ $= 120 \cos \theta \times \left(-\frac{\pi}{60} \right)$ $\frac{d}{dt}(A)$ when $\theta = \pi/6$ $= 120 \times \cos \frac{\pi}{6} \times \left(-\frac{\pi}{60} \right)$ $= 120 \times \frac{\sqrt{3}}{2} \times -\frac{\pi}{60}$ $= -\pi\sqrt{3} \text{ cm}^2/\text{min}$ $\therefore$ The area is decreasing at the rate of $\pi\sqrt{3} \text{ cm}^2$ per minute.	✓	$\approx 1 - \left(\frac{55}{60} \times \frac{54}{59} \times \frac{53}{58} \right)$ $\approx 1 - \frac{55c_3}{60c_3}$ $\approx 1 - \frac{57c_3}{60c_3}$ etc.	

MARKERS: Q1: DS Q2: EH Q3: lm Q4: RD Q5: AJ Q6: DS Q7: AJ

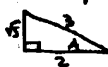
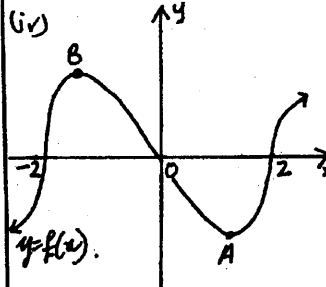
Year 12-2005 Trial HSC Mathematics EXTENSION 1 Assessment Task 4
Suggested Solutions and Marking Scheme

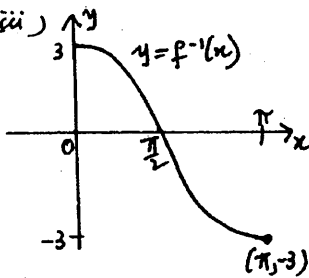
Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
QUESTION 1: (12 MARKS)			
(a) $\frac{d}{dx} \tan^{-1} 2x = \frac{2}{1+4x^2}$	✓	$\tan \theta = \left \frac{2-1}{1+2} \right $ $\tan \theta = \frac{1}{3}$ $\therefore \theta = 18.4^\circ$	✓
(b) $k:l = 3:2$ $A(-5,12) \quad B(4,9)$ $\therefore P(x,y) = \left[\frac{kx_2 + lx_1}{k+l}, \frac{ky_2 + ly_1}{k+l} \right]$ $= \left[\frac{3 \times 4 + 2 \times (-5)}{3+2}, \frac{3 \times 9 + 2 \times 12}{3+2} \right]$ $= \left[\frac{2}{5}, \frac{51}{5} \right]$	✓ ✓	(c) $\int \sin^2 3x \, dx$ N.B. $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$ $\therefore \sin^2 3x = \frac{1}{2}(1 - \cos 6x)$ $\therefore \int \sin^2 3x \, dx$ $= \frac{1}{2} \int 1 - \cos 6x \, dx$ $= \frac{1}{2} \left[x - \frac{1}{6} \sin 6x \right] + C$	✓ ✓
(c) $\lim_{x \rightarrow 0} \left(\frac{\tan 3x}{2x} \right)$ $= \lim_{x \rightarrow 0} \left(\frac{\tan 3x}{3x} \right) \times \left(\frac{3x}{2x} \right)$ $= \frac{3}{2} \times 1$ $= \frac{3}{2}$	✓	(f) Let $u = x-1$ N.B. $\frac{dx}{du} = 1$ when $x=2$, $u=1$ $x=5$, $u=4$. Also $x = u+1$ from (i) $\therefore \int_2^5 \frac{x}{\sqrt{x-1}} \, dx$ $= \int_1^4 \frac{u+1}{\sqrt{u}} \, du$ $= \int_1^4 (u^{1/2} + u^{-1/2}) \, du$ $= \left[\frac{2u^{3/2}}{3} + 2u^{1/2} \right]_1^4$ $= \left[\frac{16}{3} + 4 - \frac{2}{3} - 2 \right]$ $= \frac{10}{3}$	✓ ✓
(d) $y_1 = x^2 \therefore y_1' = 2x$ $y_2 = x \therefore y_2' = 1$ when $x=1$, $y_1' = 2$ $y_2' = 1$ Let $\theta =$ acute angle $\therefore \tan \theta = \left \frac{m_1 - m_2}{1 + m_1 m_2} \right $	✓		

Year 12-2005 Trial HSC Mathematics EXTENSION 1 Assessment Task 4
Suggested Solutions and Marking Scheme

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
QUESTION 2: (12 MARKS)			
(a)(i) $\alpha + \beta + \gamma = 5$ (ii) $\alpha\beta + \alpha\gamma + \beta\gamma = -2$ (iii) $\alpha^2 + \beta^2 + \gamma^2$ $= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$ $= 5^2 - 2 \times (-2)$ $= 25 + 4$ $= 29$	✓ ✓ ✓ only if includes $2\alpha\beta$	$y' = \pi \int_0^{\frac{1}{\sqrt{2}}} \frac{1}{\sqrt{9(1-x^2)}} \, dx$ $= \frac{\pi}{3} \int_0^{\frac{1}{\sqrt{2}}} \frac{1}{\sqrt{1-x^2}} \, dx$ $= \frac{\pi}{3} [\sin^{-1} 3x]_0^{\frac{1}{\sqrt{2}}}$ $= \frac{\pi}{3} [\sin^{-1} \frac{1}{\sqrt{2}} - \sin^{-1} 0]$ $= \frac{\pi}{3} \times \frac{\pi}{4}$ $V = \frac{\pi^2}{12}$ cubic units.	✓ ✓
(b) Let $f(x) = \ln x - \sin x$ $f(2) = \ln 2 - \sin 2$ $= -0.216 \dots$ $f(2.5) = \ln 2.5 - \sin 2.5$ $= 0.3178 \dots$ > 0 since $ -0.216 \dots < 0.3178 \dots $ then we conclude the root lies closer to $x=2$. or consider $f\left(\frac{2+2.5}{2}\right) = f(2.25)$ $= 0.0328$ > 0 $\therefore$ The desired interval is $\{x: 2 < x < 2.25\}$ hence, the root lies closer to $x=2$.	ln radians. ✓ Conclusion	(d)(i) If $5^x = e^{kx}$ then $k = \log_e 5$. (ii) $\frac{d}{dx} (5^x)$ $= \frac{d}{dx} e^{(\ln 5)x}$ $= \ln 5 \cdot e^{(\ln 5)x}$ $= \ln 5 \cdot (5^x)$ (or $\ln 5 \cdot e^{x \ln 5}$)	will not accept $\frac{\ln 5^x}{x}$ but $\frac{\ln 5}{\ln e}$ is OK.
(c) $V = \pi \int_0^{\frac{1}{\sqrt{2}}} \left[\frac{1}{\sqrt{1-9x^2}} \right]^2 dx$ $V = \pi \int_0^{\frac{1}{\sqrt{2}}} \frac{1}{1-9x^2} \, dx$	✓ only if correct formula applied		

the statement $S(n)$ is also true for all positive integer values of $n \geq 1$.

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>Q3 ctd.</u> (c) Let $A = \cos^{-1} \frac{2}{3}$ $B = \sin^{-1} \frac{1}{4}$ $\therefore \cos A = \frac{2}{3}$  $\sin B = \frac{1}{4}$  $\therefore \sin(A+B)$ $= \sin A \cos B + \sin B \cos A$ $= \frac{\sqrt{5}}{3} \times \frac{\sqrt{15}}{4} + \frac{1}{4} \times \frac{2}{3}$ $= \frac{5\sqrt{3}}{12} + \frac{2}{12}$ $= \frac{5\sqrt{3}}{12} + \frac{1}{6}$ (d) (i) PERIOD = 3 (ii) N.B: $f(x) = \begin{cases} x & 0 \leq x \leq 2 \\ 6-2x & 2 \leq x \leq 3 \end{cases}$ $\therefore f(32) = f(1)$ $= 1$ (iii) $f'(26\frac{1}{2}) = f'(2\frac{1}{2})$ $= -2$	✓ or implied ✓ ✓ ✓	<u>QUESTION 4: (12 MARKS)</u> (a)(i) $f(x) = x \left[\sqrt{x^2 - 4} \right]$ $f(-x) = (-x) \left[\sqrt{(-x)^2 - 4} \right]$ $= -x \left[\sqrt{x^2 - 4} \right]$ $= -f(x)$ $\therefore f(x)$ is an odd fn in x about $x=0$. (ii) Since $f(x)$ is an odd fn about $(0,0)$ and one of the S.P's is at $A \left[2\sqrt{\frac{3}{5}}, -\frac{4\sqrt{3}}{5} \right]$ then the other S.P is at: $B \left[-2\sqrt{\frac{3}{5}}, \frac{4\sqrt{3}}{5} \right]$ (iii) Among the reasons, one could include: $f'(x)$ is undefined when $x=2$ - the slope of the curve is "vertical" (iv) 	✓ ✓ must be stated. ✓ ✓ Vertical feature and correct shape

Suggested Solution (a)	Comments	Suggested Solution (a)	Comments
Q4 (b): (b) $f(x) = \cos^{-1}\left(\frac{x}{3}\right)$ here: $D: \left \frac{x}{3}\right \leq 1$ $\therefore x \leq 3$ $\& R: 0 \leq y \leq \pi$. (i) we seek $f^{-1}(x)$ such that: $f(f^{-1}(x)) = x$ $\therefore \cos^{-1}\left[\frac{f^{-1}(x)}{3}\right] = x$ $\therefore \cos x = \frac{f^{-1}(x)}{3}$ $\therefore f^{-1}(x) = 3 \cos x$. (ii) $D: 0 \leq x \leq \pi$ $R: y \leq 3$ (iii) 	✓ ✓ ✓ of equivalent ✓ correct shape, position, and intercept	(c) (i) $v^2 = u^2 (a^2 - x^2)$ Let $v=0, a=10, x=-8$ $\therefore 0 = u^2 (100 - 64)$ ✓ $\frac{9}{36} = u^2$ $\therefore u = \frac{3}{6} = \frac{1}{2} \text{ (m/s)}$ $\therefore T = \frac{2\pi}{u} = \frac{2\pi}{\frac{1}{2}}$ ✓ $= 4\pi \text{ seconds}$ (ii) $v=0, a=10, u=\frac{1}{2}$ $\therefore 0 = \frac{1}{4} (100 - x^2)$	

$\therefore |x| = 10 \Rightarrow x = \pm 10$ m. ✓
(both must be stated).

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>QUESTION 5: (2 MARKS)</u> (a) $\ddot{x} = 4(1+x)$ when $t=0$, $x=0$, $\dot{x}=2$ (i) $LHS = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$ $= \frac{1}{2} \times \frac{d}{dx} v^2$ $= \frac{1}{2} \times 2v \frac{dv}{dx}$ $= v \frac{dv}{dx}$ $= \frac{dx}{dt} \times \frac{dv}{dx}$ $= \frac{dv}{dt}$ $= \ddot{x}$ $= RHS.$ alternatively one could should show $RHS=LHS.$ (ii) Let $\frac{d}{dx} \frac{1}{2} v^2 = 4(1+x)$ $\therefore \int \frac{d}{dx} \frac{1}{2} v^2 dx = 4 \int (1+x) dx$ $\therefore \frac{1}{2} v^2 = \frac{4(1+x)^2}{2} + C_1$ When $x=0$, $v=2$ $\therefore 2 = \frac{4 \times 1}{2} + C_1$ $\therefore C_1 = 0$ $\therefore v^2 = 4(x+1)^2$ $\therefore v = \pm 2(x+1).$ We are told $v > 0$ $\therefore$ let $v = 2(x+1)$	✓ ✓ ✓ ✓	$\therefore \frac{dx}{dt} = 2(x+1)$ $\therefore \frac{dt}{dx} = \frac{1}{2(x+1)}$ $\therefore t = \frac{1}{2} \ln(x+1) + C_2$ When $x=0$, $t=0$ $\therefore 0 = \frac{1}{2} \ln 1 + C_2$ $\therefore C_2 = 0$ $\therefore t = \frac{1}{2} \ln(x+1)$ $\therefore e^{2t} = x+1$ $\therefore x = e^{2t} - 1$ as required. (b)(i) We know: $\frac{dM}{dt} = k(100-M).$ Suppose $M = 100 + Ae^{-kt}$ ① then $\frac{dM}{dt} = 0 - A k e^{-kt}$ $= -k(Ae^{-kt})$ From ① $M - 100 = Ae^{-kt}$ $\therefore \frac{dM}{dt} = -k(M - 100)$ $\therefore \frac{dM}{dt} = k(100 - M)$ We know $\frac{dM}{dt} = k(100 - M)$ then $\frac{dM}{dt} = k[100 - (100 + Ae^{-kt})]$ $= k[-Ae^{-kt}]$ $= -k(M - 100)$ as expected.	✓ ✓ ✓ ✓